

Latches

- ⇒ Latches are low level serialization mechanism used to protect shared data structures in the SGA from concurrent access.
- ⇒ They ensure data integrity and data consistency of memory data structures by preventing multiple processes from simultaneously modifying the same memory area.
- ⇒ Latches are points of concurrency. ⇒ Concurrent modification of data structure in following situation.

Characteristics of Locks:

- Lightweight
- Uses spin locks mechanisms
- Latch hold short time
- Deallocation of memory while being acquired.

Common types of latches:

- Shared pool latch
- Library cache latch
- Cache buffers chains latch
- Redo allocation latch
- Redo copy latch
- Checkpoint queue latch

of all latches.

⇒ V\$LATCHHOLDER: Display current latch holder.

⇒ V\$LATCHNAME: Contains name address of all latches.

DATE: / / 20

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Topic:

Types of Locks

⇒ Locks are mechanisms used to manage concurrent access to resources and ensure data consistency.

DDL Locks

Row Locks (Tx):

⇒ These locks are reserved on rows when performing DML operations like INSERT, UPDATE, DELETE, and SELECT ... FOR UPDATE.

⇒ They prevent other transactions from modifying the same rows until the transaction is committed or rolled back.

Table Locks (TM):

⇒ These locks are placed on the tables to control DML on tables. Table locks come in several modes.

Row Share (RS):

Allows concurrent access to table but prevents exclusive table-level locks.

Row Exclusive (RX):

Prevents other transactions from locking the table in exclusive mode.

Share (S):

Allows concurrent queries but prevents DML operations.

Share Row Exclusive (SRX):

Allows queries and DML but prevents other transactions from obtaining the same type of lock.

Exclusive (X):

Prevents all other access to the table until the transaction is complete.

DATE: ___/___/20__

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DDL Lock

Shared (S) Locks:

⇒ Acquired during DDL operations to ensure that the structure of the object ^(table) cannot be modified while the operation is in progress.

⇒ CREATE INDEX operation will acquire a shared lock.

Exclusive (X) Locks:

Acquired during DDL operations that modify the structure of object and prevent others.

Internal Locks

Latches:

⇒ Lightweight, low-level used to protect shared memory structures within the SGA.

⇒ Ensures data integrity and consistency in-memory data structures.

mutexes

⇒ Used to protect memory structures.

⇒ Provides finer-grained locking compared to latches.

⇒ Reduce contention.

Distributed LocksDistributed transaction locks (DLTs)

⇒ Used in distributed transactions to ensure consistency across multiple databases.

⇒ Oracle's two-phase commit protocol utilizes these locks to manage transactions that span multiple databases.

Cluster LocksGlobal engine services (GES)

⇒ GES manages engine lock across instances to ensure global data integrity and consistency.

Global cache services (GCS)

⇒ Manage cache coherency across instances by ensuring

that each data block is consistent across all clusters.

Advisory LocksUser-defined locks

⇒ Using the 'DBMS_LOCK' package, users can create their own locks to manage the application-specific resources.

⇒ Advisory in nature and do not enforce any constraints.

File and log locksFile locks

Used to manage access to data files and control files integrity.

Redo log locks

manage access to redo log files to ensure data is written correctly and in order.

DATE: 1/1/20

MTWTFSS

Configuring Archive log mode

⇒ Configuring archive log mode is essential for enabling the database to archive the redo log files, which are crucial for backup and recovery operation configuration include:

- enabling
- Disabling

Enabling archive log mode

- Shut down the database
 - SHUTDOWN IMMEDIATE;
- mount the database
 - STARTUP MOUNT;

Enable archive log mode

- ALTER DATABASE ARCHIVELOG;

DATE: 1/1/20

MTWTFSS

4. Open the database:

- ALTER DATABASE OPEN;

5. verify the database change:

- SELECT LOG-MODE FROM V\$DATABASE;

Note:

⇒ follow all these steps for disabling just change:

- ALTER DATABASE ~~NONARCHIVELOG~~;

Configuring Archive destination

ALTER SYSTEM SET LOG_ARCHIVE_DEST = '/path...';

Additional configuration Parameters

1. LOG_ARCHIVE_FORMAT:

- Specifies the format for archive log files.

o ALTER SYSTEM SET LOG_ARCHIVE-
FORMAT='arch-%t-%s'
DATE: __/__/20 MTWTFSS

2. LOG_ARCHIVE_MAX_PROCESSES:

⇒ Specifies the minimum number
of archived processes (ARCH).

o ALTER SYSTEM SET LOG_ARCHIVE
MAX_PROCESSES=4;

DATE: __/__/20 MTWTFSS

Q. What do you mean by
block efficiency?

⇒ Block efficiency refers to
how effectively the database
manage and utilizes its data
blocks in memory and on
disk.

⇒ measure of how efficiently
Oracle stores and retrieves data,
aiming to minimize I/O operations
and maximize performance.

⇒ By carefully managing block
size, compression, indexing, buffer
cache, and storage configurations,
Oracle can achieve optimal
block efficiency.

Q. List different tuning tools?

- o Oracle enterprise manager (OEM)
- o Automatic workload repository (AWR)
- o Automatic database diagnostic monitor (ADDM)
- o SQL Tuning Advisor
- o Data performance analyzer
- o TKPROF & Oracle trace

Q. How to avoid dynamic allocation

⇒ It refers to automatic extension of data files or tablespaces when they reach their maximum sizes.

⇒ it can be convenient, but can lead to:

- fragmentation
- Performance issues
- Potential down time.

Some strategies to avoid dynamic allocation:

- Proper initial sizing
- Regular capacity planning
- Preallocate space
- DB compression
- Regular database reorganization
- Capacity planning tools